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Circuitry for medical stimulation systems

Abstract

A medical device that includes a flexible body having a proximal end and a distal end, and an electrode positioned on the body proximate the distal end. The electrode is configured to provide an electrical charge for stimulating tissue. The medical device includes an electrical connection positioned on the body proximate the proximal end. The electrical connection is configured to electrically couple the electrode to a power source. The medical device includes an electrical lead connecting the electrode to the electrical connection. The lead is on or in the flexible body. The body is configured to have a first configuration prior to being secured to an exterior of a tube and a second configuration having a shape that conforms to a profile of the tube with the electrode secured to an exterior of the tube.

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8554323	12/2012	Haefner et al.	N/A	N/A
8560072	12/2012	Caparso et al.	N/A	N/A
8560086	12/2012	Just et al.	N/A	N/A
8571662	12/2012	Hoffer	N/A	N/A
8571685	12/2012	Daglow et al.	N/A	N/A
8615297	12/2012	Sathaye et al.	N/A	N/A
8617228	12/2012	Wittenberger et al.	N/A	N/A
8620412	12/2012	Griffiths et al.	N/A	N/A
8620450	12/2012	Tockman et al.	N/A	N/A
8626292	12/2013	Mccabe et al.	N/A	N/A
8630707	12/2013	Zhao et al.	N/A	N/A
8644939	12/2013	Wilson et al.	N/A	N/A
8644952	12/2013	Desai et al.	N/A	N/A
8646172	12/2013	Kuzma et al.	N/A	N/A
8650747	12/2013	Kuzma et al.	N/A	N/A
8676323	12/2013	Ignagni et al.	N/A	N/A
8676344	12/2013	Desai et al.	N/A	N/A
8694123	12/2013	Wahlstrand et al.	N/A	N/A
8696656	12/2013	Abboud et al.	N/A	N/A
8706223	12/2013	Zhou et al.	N/A	N/A
8706235	12/2013	Karamanoglu et al.	N/A	N/A
8706236	12/2013	Ignagni et al.	N/A	N/A
8718763	12/2013	Zhou et al.	N/A	N/A
8725259	12/2013	Kornet et al.	N/A	N/A
8738154	12/2013	Zdeblick et al.	N/A	N/A
8755889	12/2013	Scheiner	N/A	N/A
8774907	12/2013	Rothenberg	N/A	N/A
8781578	12/2013	Mccabe et al.	N/A	N/A
8781582	12/2013	Ziegler et al.	N/A	N/A
8781583	12/2013	Cornelussen et al.	N/A	N/A
8801693	12/2013	He et al.	N/A	N/A
8805511	12/2013	Karamanoglu et al.	N/A	N/A
8838245	12/2013	Lin et al.	N/A	N/A
8858455	12/2013	Rothenberg	N/A	N/A
8863742	12/2013	Blomquist et al.	N/A	N/A
8886277	12/2013	Kim et al.	N/A	N/A
8897879	12/2013	Karamanoglu et al.	N/A	N/A
8903507	12/2013	Desai et al.	N/A	N/A
8903509	12/2013	Tockman et al.	N/A	N/A
8909341	12/2013	Gelfand et al.	N/A	N/A
8914113	12/2013	Zhang et al.	N/A	N/A
8918169	12/2013	Kassab et al.	N/A	N/A
8918987	12/2013	Kuzma et al.	N/A	N/A
8923971	12/2013	Haefner et al.	N/A	N/A
8942823	12/2014	Desai et al.	N/A	N/A

8942824	12/2014	Yoo et al.	N/A	N/A
8948884	12/2014	Ramachandran et al.	N/A	N/A
8968299	12/2014	Kauphusman et al.	N/A	N/A
8972015	12/2014	Stack et al.	N/A	N/A
8983602	12/2014	Sathaye et al.	N/A	N/A
9008775	12/2014	Sathaye et al.	N/A	N/A
9026231	12/2014	Hoffer	N/A	N/A
9037264	12/2014	Just et al.	N/A	N/A
9042981	12/2014	Yoo et al.	N/A	N/A
9072864	12/2014	Putz	N/A	N/A
9072896	12/2014	Dar et al.	N/A	N/A
9072899	12/2014	Nickloes	N/A	N/A
9108058	12/2014	Hoffer	N/A	N/A
9108059	12/2014	Hoffer	N/A	N/A
9125578	12/2014	Grunwald	N/A	N/A
9138580	12/2014	Ignagni et al.	N/A	N/A
9138585	12/2014	Saha et al.	N/A	N/A
9144680	12/2014	Kaula et al.	N/A	N/A
9149642	12/2014	Mccabe et al.	N/A	N/A
9168377	12/2014	Hoffer	N/A	N/A
9174046	12/2014	Francois et al.	N/A	N/A
9199075	12/2014	Westlund	N/A	N/A
9205258	12/2014	Simon et al.	N/A	N/A
9216291	12/2014	Lee et al.	N/A	N/A
9220898	12/2014	Hoffer	N/A	N/A
9226688	12/2015	Jacobsen et al.	N/A	N/A
9226689	12/2015	Jacobsen et al.	N/A	N/A
9242088	12/2015	Thakkar et al.	N/A	N/A
9259573	12/2015	Tehrani et al.	N/A	N/A
9295846	12/2015	Westlund et al.	N/A	N/A
9314618	12/2015	Imran et al.	N/A	N/A
9333363	12/2015	Hoffer et al.	N/A	N/A
9345422	12/2015	Rothenberg	N/A	N/A
9370657	12/2015	Tehrani et al.	N/A	N/A
9398931	12/2015	Wittenberger et al.	N/A	N/A
9415188	12/2015	He et al.	N/A	N/A
9427566	12/2015	Reed et al.	N/A	N/A
9427588	12/2015	Sathaye et al.	N/A	N/A
9468755	12/2015	Westlund et al.	N/A	N/A
9474894	12/2015	Mercanzini et al.	N/A	N/A
9485873	12/2015	Shah et al.	N/A	N/A
9498625	12/2015	Bauer et al.	N/A	N/A
9498631	12/2015	Demmer et al.	N/A	N/A
9504837	12/2015	Demmer et al.	N/A	N/A
9526449	12/2015	Shaker	N/A	N/A
9532724	12/2016	Grunwald et al.	N/A	N/A
9533160	12/2016	Brooke et al.	N/A	N/A
9539429	12/2016	Brooke et al.	N/A	N/A
9545511	12/2016	Thakkar et al.	N/A	N/A
9561369	12/2016	Burnes et al.	N/A	N/A

9566436	12/2016	Hoffer et al.	N/A	N/A
9572982	12/2016	Burnes et al.	N/A	N/A
9597509	12/2016	Hoffer et al.	N/A	N/A
9615759	12/2016	Hurezan et al.	N/A	N/A
9623239	12/2016	Francois et al.	N/A	N/A
9623252	12/2016	Sathaye et al.	N/A	N/A
9662494	12/2016	Young et al.	N/A	N/A
9682235	12/2016	O'Mahony et al.	N/A	N/A
9694185	12/2016	Bauer	N/A	N/A
9717899	12/2016	Kuzma et al.	N/A	N/A
9724018	12/2016	Cho et al.	N/A	N/A
9744349	12/2016	Westlund et al.	N/A	N/A
9744351	12/2016	Gelfand et al.	N/A	N/A
9776005	12/2016	Meyyappan et al.	N/A	N/A
9833623	12/2016	Gnanashanmugam et al.	N/A	N/A
9861817	12/2017	Cho et al.	N/A	N/A
9872989	12/2017	Jung et al.	N/A	N/A
9884178	12/2017	Bouton et al.	N/A	N/A
9884179	12/2017	Bouton et al.	N/A	N/A
9919149	12/2017	Imran et al.	N/A	N/A
9931504	12/2017	Thakkar et al.	N/A	N/A
9950167	12/2017	Hoffer et al.	N/A	N/A
9956132	12/2017	Francois et al.	N/A	N/A
9956396	12/2017	Young et al.	N/A	N/A
9968785	12/2017	Hoffer et al.	N/A	N/A
9968786	12/2017	Bauer et al.	N/A	N/A
9987488	12/2017	Gelfand et al.	N/A	N/A
9999768	12/2017	Gelfand et al.	N/A	N/A
10022546	12/2017	Hoffer	N/A	N/A
10035017	12/2017	Thakkar et al.	N/A	N/A
10039920	12/2017	Thakkar	N/A	A61B 5/287
10195429	12/2018	Thakkar et al.	N/A	N/A
10293164	12/2018	Nash et al.	N/A	N/A
10300270	12/2018	Gelfand et al.	N/A	N/A
10315035	12/2018	Bauer	N/A	N/A
10335592	12/2018	Bauer et al.	N/A	N/A
10369361	12/2018	Bauer et al.	N/A	N/A
10391314	12/2018	Hoffer et al.	N/A	N/A
10406366	12/2018	Westlund et al.	N/A	N/A
10406367	12/2018	Meyyappan et al.	N/A	N/A
10413203	12/2018	Saha et al.	N/A	N/A
10448995	12/2018	Olson	N/A	N/A
10493271	12/2018	Bauer et al.	N/A	N/A
10512772	12/2018	Hoffer et al.	N/A	N/A
10966738	12/2020	Humayun	N/A	N/A
11045614	12/2020	Andreas et al.	N/A	N/A
11122972	12/2020	Cohen et al.	N/A	N/A
11154229	12/2020	Das et al.	N/A	N/A

11154547	12/2020	Bright et al.	N/A	N/A
11161306	12/2020	Kidmose et al.	N/A	N/A
11172859	12/2020	Connor	N/A	N/A
11179541	12/2020	Metzger	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 16/898,157, filed on Jun. 10, 2020, which claims priority to U.S. Provisional Application No. 62/860,452, filed on Jun. 12, 2019, each of which is hereby incorporated by reference in its entirety. All publications, patent applications, and patents mentioned in this specification are herein incorporated by reference in their entirety to the same extent as if each individual document was specifically individually indicated to be incorporated by reference.

TECHNICAL FIELD

(1) Embodiments of this disclosure relate to systems, devices, and apparatuses for providing

electrical circuits, leads, and/or electrodes on medical devices configured for stimulation of a body. Embodiments of this disclosure generally relate to medical devices (including systems) for the stimulation of nerves and/or muscles, including a flexible circuitry construction for use during the stimulation of muscles and/or nerves. In embodiments, systems may restore, enhance, and/or modulate diminished neurophysiological functions using electrical stimulation. Some embodiments provide a flexible layer disposed over a medical device for positioning electrodes external of the device and adjacent to one or more target nerves during a procedure.

BACKGROUND

(2) Critical care patients, particularly those requiring invasive mechanical ventilation (MV), may generally experience higher levels of diaphragm, lung, brain, heart, and/or other organ injury. The respiratory muscles (e.g., diaphragm, sternocleidomastoid, scalenes, pectoralis minor, external intercostals, internal intercostals, abdominals, quadratus, etc.) may rapidly lose mass and strength during MV. A patient's lungs may suffer from ventilator-induced trauma, including high and low pressure injuries. Cognitive effects of MV may be caused by several factors, including aberrant neuro-signaling and inflammatory responses. Limiting a duration for which patients are subjected to MV may contribute toward minimizing such negative side effects. However, rapid respiratory muscle atrophy in MV patients makes it challenging to transition patients away from a dependency on MV. Options may be limited for strengthening the respiratory muscles of critical care patients, particularly for those that are on MV, so that they may regain an ability to breathe without external respiratory support.

SUMMARY

(3) Embodiments of the present disclosure relate to, among other things, systems, devices, and methods for applying stimulation to one or more anatomical targets. Embodiments of flexible circuitry apparatuses for the systems and devices described herein may be used with MV, such as, for example, stimulation of respiratory nerves and/or respiratory muscles. Each of the embodiments disclosed herein may include one or more of the features described in connection with any of the other disclosed embodiments.

(4) In one example, a medical device may include a flexible body having a proximal end and a distal end and an electrode positioned on the body proximate the distal end. The electrode is configured to provide an electrical charge for stimulating tissue. The medical device may include an electrical connection positioned on the body proximate the proximal end. The electrical connection is configured to electrically couple the electrode to a power source. The medical device may include an electrical lead connecting the electrode to the electrical connection, wherein the lead is on or in the flexible body. The body is configured to have a first configuration prior to being secured to an exterior of a tube and a second configuration having a shape that conforms to a profile of the tube with the electrode secured to an exterior of the tube

(5) Any of the systems or methods disclosed herein may include any of the following features. The medical device may include the tube. The flexible body is helically wound about the exterior of the tube in a plurality of winds, to achieve the second configuration. The medical device may include the tube, and securing the flexible body about the exterior of the tube causes the tube to have regions of varying flexibility at locations of the tube having the flexible body. The flexible body includes a dielectric substrate film disposed between the electrode and the electrical lead. The flexible body further includes a hole through the dielectric substrate film, and the hole is filled with a conductive material electrically connecting the electrode to the electrical lead. The medical device includes a sensor, an electronic chip, or an integrated circuit disposed within the dielectric substrate film and between the electrode and the electrical lead. The flexible body, in the first configuration, includes at least one segment having an enlarged width relative to an adjacent portion of the flexible body, wherein the at least one segment is less flexible than the adjacent portion of the flexible body. The flexible body, in the first configuration, includes at least one segment having a nonlinear configuration adjacent to a portion of the flexible body having a linear

configuration, wherein the at least one segment is more flexible than the portion. The medical device includes the tube. The flexible body, in the first configuration, includes at least one bend between the proximal end and the distal end such that a longitudinal axis of a proximal portion proximal to the bend is transverse to a longitudinal axis of a distal portion distal to the bend. A flexibility of a segment of the medical device having the proximal portion differs from a flexibility of a segment of the medical device having the distal portion, due to the arrangements of the proximal and distal portions on the tube. The at least one bend causes a change in direction of a helical wind of the flexible body about the tube, when the flexible body is on the second configuration. The flexible body includes at least one segment having a stiffening structure that is configured to increase a stiffness of the flexible body relative to an adjacent portion of the flexible body. The flexible body includes at least one segment at the distal end having a plurality of branches, wherein each of the plurality of branches includes an array of electrodes. The at least one segment is angled relative to the proximal end such that a longitudinal axis of the plurality of branches is transverse to a longitudinal axis of the flexible body. The flexible body includes a second segment proximal of the distal end and having a plurality of second branches, wherein each of the plurality of second branches includes a second array of electrodes.

(6) In another example, a medical device may include a tube having a longitudinal length and at least one lumen. The tube is configured to receive at least one of a device and a fluid in the at least one lumen. The medical device may include a flexible body having a stimulation array configured to provide an electrical charge for stimulating tissue. The flexible body includes a proximal portion and a distal portion separated from the proximal portion by at least one bend. The flexible body is configured to cover at least a portion of the longitudinal length of the tube such that the stimulation array is disposed about an exterior of the tubular body. The at least one bend is configured to arrange one of the proximal portion and the distal portion in a linear configuration and the other of the proximal portion and the distal portion in a nonlinear configuration relative to the tube, when the flexible body is wound about the exterior of the tube.

(7) Any of the systems or methods disclosed herein may include any of the following features. The stimulation array includes a distal electrode array positioned on the flexible body proximate to the distal portion. The distal electrode array is configured to provide the electrical charge for stimulating tissue at a first location. The stimulation array includes a proximal electrode array positioned on the flexible body proximate to the proximal portion. The proximal electrode array is configured to provide the electrical charge for stimulating tissue at a second location that is different than the first location. The stimulation array includes an electrical connection positioned on the flexible body proximate to the proximal portion and configured to electrically couple the distal electrode array and the proximal electrode array to a power source. The at least one bend causes a change in direction of a helical wind of the flexible body about the tube, and a longitudinal axis of the proximal portion proximal to the at least one bend is transverse to a longitudinal axis of the distal portion distal to the at least one bend. A flexibility of a portion of the medical device having the proximal portion differs from a flexibility of a portion of the medical device having the distal portion, due to the arrangements of the proximal portion and distal portion on the tube.

(8) In another example, a medical device may include a tube and a circuit having a flexible body including a proximal portion and a distal portion that is angled relative to the proximal portion. The distal portion having a plurality of branches each including an array of electrodes configured to provide an electrical charge for stimulating tissue. The flexible body is configured to engage the tube with the proximal portion disposed over the tube in a helical configuration, and the distal portion is disposed over the tube in a linear configuration, such that the plurality of branches is arranged substantially parallel to a longitudinal axis of the tube.

(9) Any of the systems or methods disclosed herein may include any of the following features. The plurality of branches are positioned about a circumference of the tube at spaced intervals. The circuit is configured to modify a flexibility of the tube when the flexible body engages the tube,

with the flexibility varying between a segment of the tube having the proximal portion and a segment of the tube having the distal portion.

(10) It may be understood that both the foregoing general description and the following detail description are exemplary and explanatory only and are not restriction of the invention, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate non-limiting embodiments of the present disclosure and together with the description serve to explain the principles of the disclosure.
- (2) FIG. 1 illustrates a side view of an exemplary circuit device including a series of electrical and conductive components, according to various embodiments of the present disclosure;
- (3) FIG. 2 illustrates a partial side view of the circuit device of FIG. 1 including electrical circuitry associated with the electrical and conductive components, according to one or more embodiments of the present disclosure;
- (4) FIG. 3 illustrates a perspective view of the circuit device of FIG. 1 attached to a medical device in a helical configuration, according to various embodiments;
- (5) FIG. 4A illustrates a cross-sectional view of the circuit device of FIG. 1 attached to the medical device of FIG. 3 and including an electrode, according to one or more embodiments;
- (6) FIG. 4B illustrates a cross-sectional view of the circuit device of FIG. 1 attached to the medical device of FIG. 3 and including a device component, according to one or more embodiments;
- (7) FIG. 5 illustrates a perspective view of the circuit device of FIG. 1 including a device component, according to one or more embodiments of the present disclosure;
- (8) FIG. 6 illustrates a cross-sectional view of the circuit device of FIG. 1 with the device component received within a slot of the medical device of FIG. 3, according to one or more embodiments of the present disclosure;
- (9) FIG. 7A illustrates a side view the circuit device of FIG. 1 printed on a planar sheet and including one or more curved bends, according to one or more embodiments of the present disclosure;
- (10) FIG. 7B illustrates a side view of the circuit device of FIG. 1 printed on a planar sheet and including one or more angular bends, according to one or more embodiments of the present disclosure;
- (11) FIG. 8 illustrates a perspective of the circuit device of FIG. 1 attached to the medical device of FIG. 3 and including a hub; according to an embodiment of the present disclosure;
- (12) FIG. 9 illustrates a partial perspective view of the circuit device of FIG. 1 electrically coupled to the hub of FIG. 8 in a first configuration, according to one or more embodiments of the present disclosure;
- (13) FIG. 10 illustrates a partial perspective view of the circuit device of FIG. 1 electrically coupled to the hub of FIG. 8 in a second configuration, according to one or more embodiments of the present disclosure;
- (14) FIG. 11 illustrates a partial perspective view of the circuit device of FIG. 1 electrically coupled to a connector assembly, according to one or more embodiments of the present disclosure;
- (15) FIG. 12 illustrates of a partial perspective view of the circuit device of FIG. 1 electrically coupled to the hub of FIG. 8 and including a control unit, according to an embodiment of the present disclosure;
- (16) FIG. 13 illustrates a side view of another exemplary circuit device including a widened segment having reduced flexibility, according to one or more embodiments of the present disclosure;

(17) FIG. 14 illustrates a side view of another exemplary circuit device including a non-linear segment having enhanced flexibility, according to one or more embodiments of the present disclosure;

(18) FIG. 15 illustrates a side view of another exemplary circuit device including angular bends, according to one or more embodiments of the present disclosure;

(19) FIG. 16 illustrates a perspective view of the circuit device of FIG. 15 attached to the medical device of FIG. 3 and including a flexible segment (low density) and a rigid segment (high density) between the angular bends to provide variable flexibility and other attributes, according to one or more embodiments of the present disclosure;

(20) FIG. 17 illustrates a perspective view of another exemplary circuit device attached to the medical device of FIG. 3 and including a distal segment extending parallel to the medical device, according to one or more embodiments of the present disclosure;

(21) FIG. 17A illustrates a partial side view of the circuit device of FIG. 17 with the distal segment including a bend, according to one or more embodiments of the present disclosure;

(22) FIG. 18 illustrates a side view of another exemplary circuit device including a series of stiffener mechanisms, according to one or more embodiments of the present disclosure;

(23) FIG. 19 illustrates a side view of another exemplary circuit device including an angled distal segment formed of a pair of branches, according to one or more embodiments of the present disclosure;

(24) FIG. 20 illustrates a side view of another exemplary circuit device including an angled distal segment formed of a plurality of branches, according to one or more embodiments of the present disclosure;

(25) FIG. 21 illustrates a perspective view of the circuit device of FIG. 20 attached to the medical device of FIG. 3, according to one or more embodiments of the present disclosure;

(26) FIG. 22 illustrates a side view of another exemplary circuit device including an angled intermediate segment formed of a pair of branches and an angled distal segment formed of a pair of branches, according to one or more embodiments of the present disclosure;

(27) FIG. 23 illustrates a side view of an exemplary circuit device including an electrical circuit at a proximal end, according to one or more embodiments of the present disclosure; and

(28) FIG. 24 illustrates a perspective view of the circuit device of FIG. 23 with the electrical circuit received within a connector assembly, according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(29) Throughout the following description, specific details are set forth to provide a more thorough understanding to persons skilled in the art. The following description of examples of the technology is not intended to be exhaustive or to limit the system to the precise forms of any example embodiment. Accordingly, the description and drawings are to be regarded in an illustrative sense, rather than a restrictive sense.

(30) All publications and patent applications mentioned in this specification are herein incorporated by reference in their entirety to the same extent as if each individual publication or patent application was specifically indicated to be incorporated by reference. For example, embodiments of the present disclosure may be used in combination with one or more systems, catheters, apparatuses, and electrodes described in U.S. Pat. Nos. 8,571,662, 9,242,088, 9,333,363, 9,776,005, 10,039,920, 10,293,164, U.S. Pat. Pub. 2015/0045810, U.S. Pat. Pub. 2019/0001126, U.S. Pat. Pub. 2019/0175908, U.S. Pat. Pub. 2019/0038894, and/or U.S. Pat. Pub. 2020/0147364; the disclosures of all of which are hereby incorporated by reference.

(31) Further aspects of the disclosures and features of example embodiments are illustrated in the appended drawings and/or described in the text of this specification and/or described in the accompanying claims. It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the

invention, as claimed.

(32) As used herein, the terms “comprises,” “comprising,” “including,” “having,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus.

(33) Additionally, the term “exemplary” is used herein in the sense of “example,” rather than “ideal.” As used herein, the term “proximal” means a direction closer to an operator and the term “distal” means a direction further from an operator. The term “approximately” or like terms (e.g., “about,” “substantially”) encompass values within 10% of the stated value.

(34) Reference will now be made in detail to examples of the present disclosure described above and illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(35) FIG. 1 shows a circuit device **100** according to an example of the present disclosure. Circuit device **100** may include a flexible body **102** having a longitudinal length defined between a distal end **104** and a proximal end **106**. Flexible body **102** may include a flexible, planar substrate defined by an exterior surface **101** and an interior surface **103** (FIG. 2). In the example, flexible body **102** may generally include a linear configuration between distal end **104** and an intermediate portion of flexible body **102** distal of proximal end **106**. Additionally, flexible body **102** may have a substantially constant and nominal width between distal end **104** and proximal end **106**.

(36) Flexible body **102** may include a bend **108** adjacent to proximal end **106** such that proximal end **106** may have a transverse alignment relative to the intermediate portion of flexible body **102** and distal end **104** (the axis of end **106** is transverse to the axis of the intermediate portion). As described further herein, bend **108** may form an angle that aligns a longitudinal axis of proximal end **106** with a central axis of a medical device. Flexible body **102** may further include a proximal tab **110** at proximal end **106** that may have a width that is relatively greater than a width of flexible body **102** at distal end **104** and/or the intermediate portion. Proximal tab **110** may be sized and/or shaped to receive one or more circuitry components thereon, such as conductive pads **112**.

(37) Flexible body **102** may be formed of a flexible material having pliability properties such that circuit device **100** may be configured to deform a shape, geometry, and/or configuration of flexible body **102**. For example, flexible body **102** may be formed of Liquid Crystal Polymer (LCP), thermoplastics, polyimide, PEEK polyester, silicone, other elastomers, and the like. Flexible body **102** may be construed of multiple layers of materials, such as, for example, electrically conductive and/or non-conductive materials. As described in further detail below, circuit device **100** may be configured to engage a medical device and flexible body **102** may be configured to conform to a profile of the medical device when attached thereto.

(38) Still referring to FIG. 1, circuit device **100** is an electrical connection device that may include a flexible printed circuit board and/or a flex ribbon circuit. For example, circuit device **100** may include flexible circuitry on flexible body **102**, such as one or more (e.g., a plurality) electrical and/or conductive components including conductive pads **112** and electrodes **114**. In the present example, circuit device **100** includes a pair of electrodes **114** patterned on exterior surface **101** adjacent to distal end **104** and a pair of conductive pads **112** patterned on exterior surface **101** adjacent to proximal end **106**. Conductive pads **112** are included on proximal tab **110**, and circuit device **100** may include at least one conductive pad **112** for each electrode **114** on flexible body **102**.

(39) Conductive pads **112** may be configured to connect electrodes **114** to an ancillary device (e.g., a programmable energy delivery system, a power source, etc.) for transmission of an electric current thereto. Electrodes **114** may be configured to treat a target site of a subject (e.g., a patient) when receiving the electrical current from the ancillary device. For example, electrodes **114** may be operable to stimulate a target site (e.g., a nerve, ganglia, etc.), deliver energy to tissue (e.g.,

thermal, electrical, etc.), and/or sense a physiologic parameter from or at the target site when distal end **104** is positioned at the target site.

(40) Still referring to FIG. **1**, electrodes **114** may form one or more stimulation arrays that are wired or wirelessly connected to one or more circuits, processors, devices, systems, applications, units, or controllers. Each electrode **114** of the stimulation array may include one or more nodes for delivering stimulation. As described in further detail herein, electrodes **114** may be supported on a medical device (e.g., an intravenous catheter, a guidewire, a stimulation lead, a needle, a probe, a lead, a patch, etc.) and may be configured to be placed on, or inserted into, a patient (e.g., a neck, torso, blood vessel, etc.) or placed on the surface of the skin. It should be understood that additional and/or fewer electrodes **114** of varying shapes and/or sizes may be included on flexible body **102** than those shown and described herein.

(41) Electrodes **114** may be positioned along exterior surface **101** in various complex arrangements relative to one another to form high current densities along flexible body **102**. It should be appreciated that flexible body **102** may be configured to permit enhanced variety in positioning electrodes **114** axially and radially along a circumference of a medical device (e.g., a catheter) when circuit device **100** is attached thereto. Stated differently, a position of electrodes **114** may not be limited by a structure of a medical device, such as, for example, a geometry of an inner lumen of the medical device, due to the incorporation of electrodes **114** on circuit device **100** in lieu of the medical device.

(42) Electrodes **114** may be configured to deliver stimulation as monopolar, bipolar, tripolar, or other multipolar electrodes. As used herein, a stimulation transmitted by electrodes **114** may refer to an electrical signal transmitted by a single electrode **114** (anode) to the ground reference (cathode) that is positioned away from the anode to generate a larger electrical field (monopolar electrical stimulation); an electrical signal transmitted from an anode electrode **114** to a cathode electrode **114** (bipolar stimulation); or an electrical signal transmitted from one or more electrodes **114** to one or more other electrodes **114** (multipolar electrical stimulation) (e.g., from a cathode to two anodes, from two cathodes to an anode, or from a cathode to three or more anodes, or from two cathodes to two anodes, etc.). As any of these electrode combinations and configurations are contemplated, for clarity when referring to one or more processes, methods, or modes of operation, such types of electrical signal transmission may all be referred to as a stimulation transmitted by one electrode **114** or a combination of electrodes **114**.

(43) Referring now to FIG. **2**, interior surface **103** of flexible body **102** is shown and a relative position of conductive pads **112** and electrodes **114** on exterior surface **101** are schematically depicted for illustrative purposes. Each conductive pad **112** may be connected to a corresponding electrode **114** via a conductive (electrical) lead **116** extending along interior surface **103**.

Conductive leads **116** may include circuit traces patterned on interior surface **103**, and conductive leads **116** may define a conductive pathway between conductive pads **112** and electrodes **114**.

(44) Conductive leads **116** may be configured to negate a necessity of including wires to connect electrodes **114** to a power source, thereby simplifying a circuitry construction of circuit device **100**. Each conductive lead **116** may include a distal end positioned adjacent to distal end **104** at a location of electrode **114**, and a proximal end positioned adjacent to proximal end **106** at a location of conductive pad **112**.

(45) It should be understood that conductive pads **112** and electrodes **114** may be patterned on exterior surface **101** opposite of conductive leads **116** on interior surface **103**. Circuit device **100** may include at least one conductive lead **116** for each corresponding pair of conductive pads **112** and electrodes **114** on flexible body **102**. An electrical current received by circuit device **100** from an ancillary device (e.g., a power source, etc.) at conductive pads **112** may be delivered to electrodes **114** via conductive leads **116**.

(46) In some embodiments, flexible body **102** may include a shape memory alloy wire (not shown) disposed along exterior surface **101** and/or interior surface **103**. The wire may be configured to

selectively manipulate a shape of flexible body **102**, such as, for example, in response to receiving a current and/or an increase in temperature (e.g., from a subject's body, etc.). As described in further detail below, the wire may be operable to adjust a shape of a medical device when circuit device **100** is attached thereto. Control of a shape and/or orientation of the medical device may allow for control of a stimulation field generated by electrodes **114**. As noted above, a shape and intensity of a stimulation field may be further controlled by a shape, size, charge, and/or arrangement of electrodes **114** relative to one another along flexible body **102**.

(47) Referring to FIG. 3, circuit device **100** is shown disposed over and attached to a medical device **10**. Medical device **10** may include a tubular body **12** having a longitudinal length defined between a distal tip **14** and a proximal tip **16**. In some embodiments, medical device **10** may include a catheter, a flexible tube, a sheath, a scope, a lead, a probe, a hypotube, a patch, a cable, and/or various other flexible instruments suitable for intravascular, percutaneous, or transcutaneous delivery. In the example, medical device **10** may include an intravenous catheter having a flexibility sufficient to traverse a subject's blood vessels, esophagus, nasal cavity, and/or various other anatomical lumens, passageways, etc.

(48) In some examples, tubular body **12** may be formed of an extruded polymer having homogenous properties (e.g., a constant thickness, etc.) between distal tip **14** and proximal tip **16** such that tubular body **12** may include a substantially consistent flexural strength along the longitudinal length. In other embodiments, circuit device **100** may be configured to increase a material thickness along certain portions of tubular body **12** and/or include one or more structures that modify the properties of tubular body **12** (e.g., increase or decrease rigidity or flexibility, affect shape memory, etc.) when flexible body **102** is attached thereto.

(49) Tubular device **12** may include one or more lumens (see FIG. 6) defined between distal tip **14** and proximal tip **16**. The one or more lumens may be sized, shaped, and configured to receive one or more devices therein, including, but not limited to, an irrigation device, an ultrasound device, an optical device, an imaging device, a sensing device, an illumination device, a guidewire, a diagnostic or therapeutic device, etc., and/or receive a fluid. Distal tip **14** may include an opening in communication with the one or more lumens of tubular body **12** such that the one or more devices received within the lumens may extend outwardly from distal tip **14** through the opening. Distal tip **14** may have an atraumatic configuration to minimize injury to tissue when traversed through a subject.

(50) Circuit device **100** may be disposed about medical device **10** and secured to an outer surface of tubular body **12**. Flexible body **102** is configured to have a first configuration prior to being secured to an exterior of tubular body **12** and a second configuration having a shape that conforms to a profile of the tubular body **12** with electrodes **114** secured to an exterior of tubular body **12**. Accordingly, flexible body **102** may be configured to conform to a geometry of tubular body **12** (e.g., cylindrical). In the example, distal end **104** may be secured to a distal portion of tubular body **12** adjacent to and proximal of distal tip **14**. Proximal end **106** may be secured to a proximal portion of tubular body **12** adjacent to and distal of proximal tip **16**.

(51) Circuit device **100** may be attached to medical device **10** at one or more locations and by various methods, including, for example, bonding flexible body **102** to an exterior of tubular body **12** (e.g., thermal bonding, thermal welding, thermal lamination, ultrasonic welding, laser welding, RF welding, adhesive bonding, solvent bonding, chemical reaction, and/or solvent bonding). It should be appreciated that a resulting flexural property of tubular body **12** may vary based on various factors, including but not limited to, the method of bonding flexible body **102** thereto, a configuration/arrangement of flexible body **102** relative to tubular body **12**, and/or a material composition of flexible body **102**. In some embodiments, circuit device **100** may be treated to modify the surface properties of flexible body **102** prior to limitation or bonding onto tubular body **12**. Examples of surface treatments may include, but are not limited to, micro-blasting, sand blasting, plasma treatment, chemical etching, filing, grinding, electroplating, thermal spraying,

physical vapor deposition, electron beam heating, and more.

(52) The resulting flexural properties of the combined assembly of medical device **10** and circuit device **100** may influence the performance of tubular body **12** during a procedure. For example, the flexural properties may be decreased in specific regions along a longitudinal length of tubular body **12** to influence where tubular body **12** may bend during a procedure, such as when navigating through anatomical structures of a subject. The flexural properties may be decreased to further reduce stress along specific regions of tubular body **12** that may be maintained in a bent configuration when positioned at a target site within a subject. In some embodiments, the flexural properties may be configured to influence a direction of bending tubular body **12**.

(53) Still referring to FIG. **3**, flexible body **102** may be disposed about tubular body **12** in various arrangements and/or configurations to position electrodes **114** at a desired location relative to tubular body **12**. It should be appreciated that electrodes **114** may be positioned at a variety of axial locations along tubular member **12**, and/or various radial positions around tubular member **12**, to achieve varying physical properties of medical device **10**. For example, flexible body **102** may be attached to tubular body **12** in a manner that may be parallel or transverse (e.g., perpendicular) to a central axis of tubular body **12**.

(54) In the present example, flexible body **102** may be attached to tubular body **12** at an angle that is transverse to a central axis of tubular body **12** such that circuit device **100** is secured to medical device **10** in a helical configuration. Flexible body **102** may be wound about the central axis of tubular body **12** in a plurality of helical winds **115** (i.e., helix turns). A pair of adjacent helical winds **115** may form a gap **117** therebetween. Electrodes **114** may be wrapped around a circumference of tubular body **12** and flexible body **102** may conform to a profile of tubular body **12**.

(55) Still referring to FIG. **3**, flexible body **102** may be helically wrapped with a substantially constant pitch and uniform gaps **117** between adjacent helical winds **115**. Accordingly, circuit device **100** may have constant flexural properties along a longitudinal length of flexible body **102**. In other embodiments, flexible body **102** may be helically wrapped to have varying pitches and/or gaps (see FIGS. **16-17**), including no gaps, to selectively adjust the flexural properties of circuit device **100** at various locations along tubular body **12**.

(56) A substantial length of tubular body **12** may be disposed underneath flexible body **102**. As described above, bend **108** may form an angle between an intermediate portion of flexible body **102** and proximal end **106** to align proximal end **106** with the central axis of tubular body **12**. Accordingly, the plurality of helical winds **115** may terminate at bend **108** such that proximal end **106** may extend substantially parallel to tubular body **12**. Stated differently, the helical configuration of flexible body **102** may extend from distal end **104** to bend **108** such that proximal portion **106** may be maintained in a non-helical (e.g., linear) configuration that is parallel to the central axis of tubular body **12**.

(57) Still referring to FIG. **3**, distal tip **14** may be positioned relatively distal of distal end **104**, and proximal tip **16** may be positioned proximal of bend **108** and relatively distal of proximal tab **110**. Accordingly, proximal tab **110** may extend proximally of proximal tip **16**. As described in greater detail below, proximal tab **110** may extend beyond proximal tip **16** for receipt within a hub **140** (see FIG. **9**).

(58) FIGS. **4A-4B** are exemplary partial cross-sectional views of FIG. **3**, taken at line **4-4**. Referring now to FIG. **4A**, an exemplary flexible circuitry construction of flexible body **102** is schematically depicted. For example, flexible body **102** may be formed of one or more electrically conductive and/or insulating materials that form conductive pathways between distal end **104** and proximal end **106**. Electrode **114** may define a first conductive layer and conductive lead **116** may define a second conductive layer. Circuit device **100** may further include a base layer **120** formed of one or more dielectric substrate films disposed between the pair of electrically conductive layers. Electrode **114** and conductive lead **116** may be affixed to base layer **120** by, for example,

depositing, adhering, and/or bonding the layers together.

(59) In some examples, base layer **120** may have a thickness ranging from about 12 microns to about 125 microns. The one or more dielectric substrate films forming base layer **120** may comprise polymer materials, such as, for example, polyimides, polyesters, liquid crystal polymer (LCP), polyethylene naphthalate, polyvinylidene fluoride, polyetherimide, various fluoropolymers (FEP), etc. In some examples, the polymer materials comprising base layer **120** may be biocompatible with tissue.

(60) Still referring to FIG. **4A**, the one or more electrically conductive materials forming electrode **114** and/or conductive lead **116** may comprise, for example, copper, gold, graphene, silver, nickel, platinum, platinum-iridium alloy, iridium oxide, titanium nitride, tungsten, alloys, and/or combinations thereof. In other embodiments, conductive lead **116** may be patterned to form various other electrical circuit elements.

(61) For example, the electrically conductive layer forming conductive lead **116** may be patterned using additive processes (e.g., spraying, sputtering, sputter coating, screen printing, stenciling, electrodeposition, electroplating, atomic layer deposition, and/or digital printing, etc.) and/or subtractive processes (e.g., chemical etching, laser etching, and/or laser machining, etc.). Stated differently, flexible body **102** may include the desired electrical circuit traces and/or other electrical circuit elements by selectively removing a portion of conductive lead **116** via one or more of the additive and/or subtractive processes described above.

(62) Still referring to FIG. **4A**, electrode **114** may be affixed to an opposing side of base layer **120** as conductive lead **116**. Further, electrode **114** may be interconnected with conductive lead **116** and through base layer **120** by a via **124**. Via **124** may be a hole formed through base layer **120** by, for example, drilling, laser cutting, and/or mechanically puncturing base layer **120** and/or the pair of opposing conductive layers (e.g., electrode **114**, conductive lead **116**). In some embodiments, circuit device **100** may include a plurality of vias **124**, each of which may be filled with a conductive material configured to connect electrode **114** and conductive lead **116** through base layer **120**.

(63) In some embodiments, an exterior surface of electrode **114** and/or conductive lead **116** may have one or more materials and/or finishes included thereon, such as, for example, via additive processing (e.g., electrodeposition, electroplating, electroless plating, sputtering, sputter coating, etc.). The one or more materials and/or finishes included on electrode **114** and/or conductive lead **116** may include, for example, nickel, silver, gold, platinum, platinum iridium, titanium nitride, and the like. The one or more materials and/or finishes may be operable to enhance a conductivity, corrosion resistance, and/or a biocompatibility of circuit device **100**.

(64) Additionally, an insulative or non-conductive substrate, layer, coating, film, and/or finish may be attached (e.g., bonded, soldered, etc.) to flexible body **102** to provide insulation of electrode **114**, conductive lead **116**, and/or base layer **120**. Such materials may include, for example, tetrafluoroethylene (TFE), polytetrafluoroethylene (PTFE), polyimide, polyetheretherketone (PEEK) polyester, silicon, and other various flexible substrates.

(65) Still referring to FIG. **4A**, a protective cover layer may be included on circuit device **100** and bonded to base layer **120** and/or conductive lead **116**. The protective cover layer may include a film and/or finishing (e.g., laminate adhesive layer, etc.) configured to protect the conductive lead **116** (and/or other electrical components) from environmental exposure to moisture, contaminants, mechanical damage, etc., during use of circuit device **100** in a procedure. In the present example, circuit device **100** may include a coverlay film **122** bonded to a surface of conductive lead **116** and base layer **120** with an adhesive **126**. Coverlay film **122** may be configured and operable to protect conductive leads **116**.

(66) Referring now to FIG. **4B**, another exemplary flexible circuitry construction of flexible body **102** is schematically depicted. In this example, flexible body **102** includes a plurality of base layers **120** (e.g., dielectric substrate films) secured to one another by at least one adhesive layer **126**

positioned between a pair of adjacent base layers **120**. Flexible body **102** may further include a plurality of electrodes **114** disposed over an upper-most base layer **120**, with each electrode **114** including a corresponding conductive lead **116** disposed under a lower-most base layer **120** (adjacent to tubular body **12**).

(67) Each electrode **114** may be interconnected with conductive lead **116** by a respective via **124** that extends through the plurality of base layers **120** disposed therebetween. Each via **124** may be filled with a conductive material configured to connect the conductive materials (e.g., electrode **114** and conductive lead **116**) to one another through the plurality of base layers **120** and adhesive layers **126** positioned therebetween. In the present example, flexible body **102** may include three base layers **120** and one coverlay film **122** disposed beneath base layers **120**.

(68) Still referring to FIG. **4B**, flexible body **102** may further include a device component **130** embedded between the plurality of base layers **120**. Device component **130** may be positioned in alignment with a center base layer **120** and between an upper-most base layer **120** and a lower-most base layer **120**. Flexible body **102** may further include one or more conductive layers **132** in contact with device component **130** and one or more (e.g., a plurality) vias **124** (filled with conductive material) connecting conductive layers **132** with corresponding conductive leads **116**. It should be understood that circuit device **100** may include a respective conductive pad **112** at proximal end **106** corresponding to each of the plurality of electrodes **114** and device component **130**.

(69) Device component **130** may include various systems, including, for example, a sensor, an electronic chip, an integrated circuit, and the like. In some embodiments, device component **130** may be fully enclosed between the plurality of base layers **120**; while in other embodiments, at least a portion of device component **130** may be exposed from the plurality of base layers **120**. In this instance, device component **130** may be accessible to a subject's tissue and/or fluids during use of circuit device **100** in a procedure. For example, device component **130** may include a sensing system configured to detect a signal (e.g., cardiac signal, impedance, temperature, flow, pressure, chemical composition, blood gas, etc.) from a subject (i.e., a patient) to facilitate determining a position of circuit device **100** therein, and/or to detect stimulation of an anatomical target based on a physical response of the subject.

(70) Although not shown in FIGS. **4A-4B**, in some embodiments, an adhesive, a polymer, or other filling material may be included to provide a smooth transition between one or more edges of flexible body **102** and a surface of tubular body **12**. By providing a smooth transition, medical device **10** and circuit device **100** may be inserted and/or removed from the subject with minimal resistance and potential for tissue trauma.

(71) In other embodiments, device component **130** may include flex detection circuitry that may be configured to detect the flexure of flexible body **102** during a procedure at various regions along the longitudinal length of tubular body **12**. Such information may guide a placement of tubular body **12** and/or provide anatomical information to an operator of medical device **10**. For example, device component **130** may include a strain circuit, a piezo circuit, and/or various other sensors configured to measure a compression and/or expansion (e.g., lateral extension, longitudinal elongation, etc.) of opposite surfaces of tubular body **12** to determine a two or three-dimensional spatial orientation (e.g., distance, rotation, etc.) of tubular body **12** relative to a subject.

(72) Additionally and/or alternatively, one or more integrated circuits may be directly attached to flexible body **102** to provide further localized smart circuitry. For example, as shown in FIG. **5**, one or more device components **130** may be attached (e.g., soldered) to interior surface **103** at various positions along flexible body **102**, such as adjacent to distal end **104**. In this instance, device components **130** extend laterally outward from interior surface **103**, thereby forming a protruding profile relative to interior surface **103**.

(73) Circuit device **100** may include a circuit trace **134** along interior surface **103** for each device component **130** included on flexible body **102**. Circuit traces **134** may interconnect device

components **130** with a corresponding conductive pad **112** at proximal end **106** (FIG. **1**). In the present example, medical device **10** may include a longitudinal slot, channel, and/or groove formed along an outer perimeter of tubular body **12** that is sized and shaped to receive the one or more device components **130** when flexible body **102** is secured thereto. The one or more slots on tubular body **102** may receive device components **130** therein to accommodate a flush engagement of flexible body **102** with tubular body **12**.

(74) Referring now to FIG. **6**, at least one slot **15** may be formed along at least one surface of tubular body **12**. Slot **15** may extend through tubular body **12** and into at least one lumen **13** of tubular body **12**. Accordingly, lumen **13** may be exposed to an exterior of tubular body **12** via slot **15**. It should be understood that tubular body **12** may include a plurality of other lumens **11** that are separate from the at least one lumen **13** aligned with slot **15**. Accordingly, the plurality of lumens **11** are not in fluid communication with lumen **13** or with one another. In other examples, two or more of the plurality of lumens **11** may be interconnected with one another and/or with lumen **13** at one or more locations.

(75) Flexible body **102** may be secured to tubular body **12** such that device component **130** may be aligned with and received in slot **15**. Thus, interior surface **103** may be flush against an exterior surface of tubular body **12**. It should be appreciated that flexible body **102** may cover and/or encapsulate a majority of tubular body **102**. Accordingly, slot **115** may be covered when flexible body **102** is attached to tubular body **12**, thereby sealing lumen **13** with interior surface **103**. Device component **130** may include, for example, a thermistor sensor, an electronic chip, a resistor, a diode, a capacitor, a strain gauge, a piezoelectric sensor, an integrated circuit, and various other electronic devices.

(76) By way of example, device component **130** may be configured to measure a temperature of a fluid, such as, for example, a fluid disposed within one of the plurality of lumens **11**, **13** and/or positioned external of medical device **10**. Circuit device **100** may be configured to measure a change in temperature based on the temperature data detected by device component **130**. In some examples, circuit device **100** may be configured to measure a flow rate and/or other characteristics of the fluid based on the temperature data sensed by device component **130**. In other examples, device component **130** may be configured to heat fluid (e.g., blood, etc.) located external of circuit device **100** and measure a temperature change of the fluid at one or more locations along a longitudinal length and/or lateral width of circuit device **100**. For example, at least one location for measuring a temperature change of the fluid may be positioned relatively distal to the heat source (e.g., distal component **130**) for determining a flow rate or other properties of the fluid.

(77) In some embodiments, as seen in FIG. **7A**, circuit device **100** may be printed on a dielectric substrate film in sheet format, such as on a planar sheet **50**. Flexible body **102** may be cut from sheet **50** in various desired sizes, shapes, and/or configurations. For example, flexible body **102** may be printed to have a substantially constant width between distal end **104** and proximal end **106**. Further, flexible body **102** may be printed to have various suitable lengths between distal end **104** and proximal end **106** in accordance with a longitudinal length of tubular body **12**.

(78) Proximal end **106** may be printed on sheet **10** with various configurations, sizes, and/or shapes. For example, proximal end **106** may be printed to include bend **108** having a desired angle. It should be appreciated that bend **108** may define a plurality of angles and/or degrees of curvature between proximal end **106** and an intermediate portion of flexible body **102**. Further, proximal end **106** may be printed to include a squared end defining proximal tab **110** (FIGS. **1-3**). In other examples, proximal end **106** may be printed to omit bend **108** entirely and/or to include a circular end defining another exemplary proximal tab **110'**.

(79) Still referring to FIG. **7A**, flexible body **102** may be printed on sheet **50** with one or more bends **105A** to accommodate various longitudinal lengths. Bends **105A** may be sized to minimize material use on sheet **50** when printing circuit device **100** thereon. For example, the one or more bends **105A** may range from about 90 degrees to about 180 degrees. In other examples, as seen in

FIG. 7B, flexible body **102** may be printed on sheet **50** with one or more bends **105B** that range from about 1 degree to about 89 degrees, such as 45 degrees. In this instance, flexible body **102** may be more easily straightened for attachment onto tubular body **102** than when circuit device **100** is printed with the one or more bends **105A** (e.g., bends having a greater angle).

(80) Referring now to FIG. 8, hub **140** may be configured and operable to receive tubular member **12** and flexible body **102** therein. In some embodiments, tubular member **12** and flexible body **102** may be coupled to hub **140**; while in other embodiments, hub **140** may be molded over tubular member **12** and flexible body **102**. Hub **140** may extend proximally from proximal tip **16** and include one or more extension tubes **150** coupled and/or molded to a proximal end of hub **140**. Hub **140** may be configured to interconnect lumens **11** (FIG. 6) with extension tubes **150**.

(81) Extension tubes **150** may include a coupling mechanism **152** (e.g., a luer connector, a female adapter, a fitting, and/or other fluid tight connector assemblies) at a distal end of each extension tube **150** for securely coupling an ancillary device (e.g., a tool, a sensor, a cable, a fiber, etc.). Extension tubes **150** may facilitate receipt of the one or more ancillary devices into lumens **11** via hub **140**. In some examples, extension tubes **150** may be configured to couple hub **140** to a fluid source for delivering a fluid (e.g., drug, therapeutic agent, etc.) to lumens **11**. Extension tubes **150** may be further configured to extract a fluid sample (e.g., blood) collected from medical device **10** and received at hub **140** via lumens **11**.

(82) In other examples, extension tubes **150** may receive an ancillary device for monitoring a pressure and/or a temperature (e.g., a sensor) and/or for positioning tubular body **12** within a patient (e.g., a guidewire). Other ancillary devices, such as, for example, optical fibers, ultrasound cameras, intravascular imagers, filters, lasers, lights, and the like may be introduced into lumens **11** through hub **140** and from extension tubes **150**.

(83) Still referring to FIG. 8, hub **140** may include a protective sheath **154** secured thereto. Protective sheath **154** may include an electrical connector **156** at a proximal end of protective sheath **154** and may define a channel housing one or more wires and/or cables. As described in further detail herein, the wires may be coupled to electrical connector **156** at a proximal end and be received within hub **140** for connection to flexible body **102** at a distal end.

(84) FIG. 9 shows hub **140** with at least a top wall of hub **140** omitted for illustrative purposes. Proximal end **106** may be received through an inlet **142** of hub **140** such that proximal tab **110** is disposed within hub **140**. Hub **140** may further include a proximal wall **144**, positioned opposite inlet **142**, and one or more outlets **146** formed through proximal wall **144**. Each extension tube **150** and protective sheath **154** may be coupled to at least one outlet **146**. In some examples, hub **140** may receive one or more wires **118** from protective sheath **154** via at least one outlet **146**.

(85) It should be appreciated that proximal hub **140** may incorporate one or more features for facilitating an electrical connection to wires **118**. The one or more wires **118** may be connected to conductive pads **112** to thereby establish electrical communication between wires **118** and electrodes **114** (e.g., via conductive leads **116**). An electrical junction between conductive pads **112** and wires **118** may be formed by various suitable attachment methods, including, for example, solder, laser weld, conductive adhesive, hot bar, interference, friction fits, and/or various other mechanical connection methods.

(86) Further, the electric junction may be reinforced with potting, overmolding, encapsulating, mechanical fasteners, and/or combinations thereof. Each wire **118** may extend through protective sheath **154** and may terminate at electrical connector **156**. Electrical connector **156** may be selectively coupled to a power source configured to supply electrodes **114** with electrical power.

(87) Still referring to FIG. 9, hub **140** may include a rigid body defining a fixed and stable interaction point between flexible body **102** and wires **118**. Accordingly, hub **140** may be configured to provide strain relief to flexible body **102** and wires **118** at the juncture between proximal tab **110** and wires **118**. In some embodiments, the rigid body of hub **140** may be formed of a thermoplastic material having dielectric properties. Thus, hub **140** may be configured to

electrically isolate the connections of conductive pads **112** with wires **118**.

(88) FIG. **10** shows another configuration of circuit device **100** in which at least an intermediate portion of flexible body **102** may be received through inlet **142** and into the respective outlet **146** coupled with protective sheath **154**. Accordingly, a portion of flexible body **102** may extend through protective sheath **154** and be received within electrical connector **156**. It should be understood that wires **118** may be omitted entirely from the present example such that proximal tab **110** may be directly coupled to the power source.

(89) Referring now to FIG. **11**, an alternative exemplary proximal tab **110'** is depicted at proximal end **106**. In the example, proximal tab **110'** may have a circular shape and may extend at an angle relative to proximal end **106**. In some embodiments, a position and/or orientation of proximal tab **110'** relative to proximal end **106** may be selectively adjustable. Proximal tab **110'** may include a plurality of apertures **111** formed therethrough and a plurality of conductive pads **112** positioned about the plurality of apertures **111**.

(90) Stated differently, each conductive pad **112** is formed about at least one aperture **111** such that conductive pads **112** may contact a device received through aperture **111**. Internal components of electrical connector **156** are depicted with an outer housing of electrical connector **156** omitted for illustrative purposes. Electrical connector **156** may include a socket **158** having a plurality of openings, and a plurality of pins **159** extending through the plurality of openings. Each of the plurality of pins **159** may have a proximal end extending proximally out of socket **158** and a distal end extending distally out of socket **158**.

(91) Still referring to FIG. **11**, the distal ends of pins **159** may be disposed within the outer housing of electrical connector **156** when in a fully assembled state for connection with proximal tab **110'**. The proximal ends of pins **159** may extend outwardly from the outer housing of electrical connector **156** when in a fully assembled state for connection with a power source. Accordingly, the proximal ends of the plurality of pins **159** may define a connector interface of electrical connector **156**.

(92) Each of the plurality of apertures **111** may be sized and shaped to receive at least one of the plurality of pins **159**. It should be appreciated that a quantity and position of apertures **111** relative to proximal tab **110'** may be based on a number and position of pins **159** on socket **158**. Circuit device **100** may establish connection with the power source in response to conductive pads **112** contacting pins **159**. In some examples, proximal tab **110'** may be soldered to socket **158** thereby securing an electrical connection between conductive pads **112** and pins **159**.

(93) In other embodiments, referring now to FIG. **12**, circuit device **100** may include an electronic component **135** mounted to flexible body **102** at proximal tab **110**. Electronic component **135** may be positioned along exterior surface **101** and encapsulated within hub **140**. Electronic component **135** may be secured to flexible body **102** (e.g., at proximal tab **110**) via various suitable configurations, including, for example, by a pin and via, an overmolding, potting, and the like. In other embodiments, electronic component **135** may be secured directly to hub **140**.

(94) Electronic component **135** may include various suitable devices, including, for example, computer readable storage media such as various types of volatile and non-volatile storage media, including but not limited to, static memory, dynamic memory, a read-only memory chip, a read-write memory chip, an electrically erasable programmable read-only memory (EEPROM), random access memory, flash memory, and the like. In other embodiments, electronic component **135** may include a sensing circuit, a radio-frequency identification (RFID) tag, and other various devices.

(95) For example, electronic component **135** may be configured to store product specific data relating to circuit device **100**, such as a serial number, lot number, product configuration, expiration date, calibration constants for device component **130**, etc. By way of further example, electronic component **135** may be configured to record use data of circuit device **100** and/or medical device **10**, such as dates of activation, variable data from therapy procedure sessions, control unit function algorithms, sensor data of device component **130**, etc.

(96) In some embodiments, electronic component **135** may further include a computer system, such as, for example a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), or both) in conjunction with the memory described above. For example, the processor may be part of a standard personal computer or a workstation. The processor may be one or more general processors, digital signal processors, application specific integrated circuits, field programmable gate arrays, servers, networks, digital circuits, analog circuits, or combinations thereof, for analyzing and processing data. The processor may implement a software program, such as programmable code. In other embodiments, the memory may be an external storage device or database separate from electronic component **135**.

(97) Still referring to FIG. **12**, electronic component **135** may include a controller that is generally configured to accept information from one or more other components of circuit device **100** (e.g., electrodes **114**, device component **130**, etc.), and process the information according to various algorithms to produce control signals for controlling the other components of circuit device **100**.

(98) For example, the controller may accept information from the system and system components, process the information according to various algorithms, and produce information signals that inform a user of circuit device **100** of a procedure status or other information that is being monitored by circuit device **100**. The controller may be a digital IC processor, analog processor, or any other suitable logic or control system that carries out control algorithms. Electronic component **135** may be powered by a battery, a direct wire energy, or other suitable power devices. In some examples, electronic component **135** may be configured to communicate with the controller or other components of circuit device **100** via a wired and/or wireless connection.

(99) According to an exemplary method of using circuit device **100** during a procedure, flexible body **102** may be initially bonded to tubular body **12** in one of a plurality of configurations as shown and described above. Proximal tab **110** may be received within hub **140** and coupled to a power source from a connection of conductive pads **114** with electrical connector **156**. With circuit device **100** secured to medical device **10**, tubular body **12** may be received within a subject (e.g., patient) and navigated toward a target treatment site.

(100) Medical device **10** may be configured to traverse through blood vessels located relatively proximate to one or more nerves (e.g., phrenic nerves, vagus nerves, etc.), muscles, and/or other tissues. A flexibility (or lack thereof) of tubular body **12** may be at least partially determined by the flexural properties of circuit device **100** based on an attachment configuration of flexible body **102** and/or a material composition of flexible body **102**. In other words, a geometry and arrangement of circuit device **100** may influence the flexural properties of medical device **10**. Medical device **10** may be guided to a target treatment site using one or more devices received within lumens **11** (FIG. **6**) via extension tubes **150** and hub **140**. For example, medical device **10** may receive a stiffening element (e.g., polymer and/or metallic mandrels), a guidewire, a sensor, an imaging device, an illumination device, a shape memory material (e.g., Nitinol, etc.), and/or other therapeutic or diagnostic medical devices to facilitate placement of circuit device **100**.

(101) Accordingly, tubular member **12** may be configured to guide placement of circuit device **100** relative to the target treatment site. Upon arriving at the target treatment site, tubular body **12** may be moved (e.g., translated, rotated, etc.) to position electrodes **114** at a desired location relative to a target object (e.g., a nerve, a muscle, a tissue, etc.). With electrodes **114** positioned adjacent to the target object, activation of the power source may deliver an electrical current through wires **118**, conductive pads **112**, and to electrodes **114** via conductive leads **116**. Electrodes **114** may be configured to stimulate the target object (e.g., a stimulation sufficient to cause a desired physiological result) upon receiving the electrical current from conductive leads **116**.

(102) Electrodes **114** may generate therapeutic effects for specific anatomical targets (e.g., nerve stimulation) and a relative position of electrodes **114** to one another on exterior surface **101** may generate one or more charge fields along tubular body **12** for certain therapeutic effects (e.g., asymmetric charge fields). Electrodes **114** may generate a sufficient charge to cause contraction of

a diaphragm or other lung-accessory. In some embodiments, electrodes **114** may be positioned along tubular body **12** to form a sensor array antenna configured and operable to detect various forms of stimulation, including, for example, nerve stimulation, muscle stimulation, cardiac stimulation, phrenic nerve activation, diaphragm muscle activation, and more.

(103) In examples where circuit device **100** includes device component **130** (e.g., a sensor, etc.), device component **130** (FIGS. **4B-6**) may be positioned toward the target object and activated to detect and/or record one or more characteristics or properties of the target object (e.g., input on whether the requisite stimulation of an anatomical target occurred). Device component **130** may be communicatively coupled to one or more computing systems, such as, for example, via electrical component **135** (FIG. **12**) by a wired or wireless connection.

(104) It should be understood that with circuit device **100** secured to an exterior of medical device **10**, lumens **11** may be utilized to receive other devices than those included in circuit device **100**. Stated differently, circuit device **100** may be operable to maximize an available space within lumens **11** to receive other ancillary devices. For example, circuit device **100** may omit a necessity to route wires internally through tubular body **12** for connecting electrodes **114** to a power source, thereby allowing tubular body **12** to be sized with a smaller minimum diameter.

(105) Further, circuit device **100** includes conductive leads **116** such that use of wires to connect electrodes **114** to a power source may be omitted entirely, whether received within and/or disposed over tubular body **12**, thereby further reducing a complexity of medical device **10**. Accordingly, tubular body **12** may include a smaller cross-sectional profile than other medical devices (e.g., catheters) sized and shaped to accommodate electrical components (e.g., wires) for including one or more electrodes on the device.

(106) In some embodiments, medical device **10** may sense physiologic parameters of the target treatment site using one or more devices (e.g., intravascular ultrasound (IVUS)) received within lumens **11** (FIG. **6**) via extension tubes **150** and hub **140**. In other embodiments, medical device **10** may receive devices that vary the flexural properties of tubular body **12** to create zones of stiff and/or flexible regions. For example, a Nitinol wire (not shown) may be received within lumen **11** to alter or control the shape of tubular body **12** in response to the wire receiving a current applied thereto. In this instance, the wire may apply a force onto tubular body **12** from within lumen **11** to a fixed/desired configuration. Removal of the current may terminate the force applied to tubular body **12** by the wire, thereby permitting tubular body **12** to return to a flexible state. In other examples, the shape memory wire may be activated by the heat produced by the subject's body.

(107) FIGS. **13-24** show alternative exemplary circuit devices that may be substantially similar to circuit device **100**. A geometry, orientation, and layer construction of the circuit devices relative to an axis of tubular body **12** may influence the flexural properties of medical device **10** in specific regions along the longitudinal length of tubular body **12**. It should be understood that the circuit devices described in the examples below may be configured and operable similar to circuit device **100** except for the differences explicitly noted herein, and therefore like reference numerals are used to identify like components. It should further be appreciated that the exemplary circuit devices below may be readily incorporated onto medical device **10**.

(108) Referring specifically to FIG. **13**, an exemplary circuit device **200** is shown in accordance with an example of this disclosure. Circuit device **200** may include a varying geometry and/or width along a longitudinal length of flexible body **102**. For example, flexible body **102** may include one or more widened segments **203** between distal end **104** and proximal end **106**. In the present example, circuit device **200** may include a pair of widened segments **203** separated from one another along the longitudinal length of flexible body **102** by an intermediate segment **205** having a relatively smaller width.

(109) Flexible body **102** may include layers having a greater thickness along widened segments **203** relative to intermediate segment **205**. Widened segments **203** may be configured to impart reduced flexibility on flexible body **102**. In the example, widened segments **203** have a relatively

smaller longitudinal length than intermediate segment **205**, however, it should be understood that various other relative lengths and/or configurations of segments **203**, **205** may be included without departing from a scope of this disclosure. In other embodiments, additional and/or fewer widened segments **203** and/or intermediate segments **205** may be included along flexible body **102**.

(110) Still referring to FIG. **13**, flexible body **102** may include a plurality of electrodes **114** along the pair of widened segments **203** and intermediate segment **205**. Thus, proximal tab **110** may include a plurality of conductive pads **112** at proximal end **106** in accordance with a quantity of electrodes **114** on flexible body **102**. In some embodiments, one or more of widened segments **203** and/or intermediate segment **205** may exclude any electrodes **114** thereon. In use, widened segments **203** may be configured and operable to reduce the flexural properties of tubular body **12** along the respective portions of tubular body **12** that align with widened segments **203** when circuit device **200** is attached to medical device **10**.

(111) Referring now to FIG. **14**, an exemplary circuit device **300** is shown in accordance with another example of this disclosure. Circuit device **300** may include an irregular segment **303** along a longitudinal length of flexible body **102**. Irregular segment **303** may be included between distal end **104** and proximal end **106** and may have a geometry and/or configuration that is transverse to a remaining linear geometry of flexible body **102**. For example, irregular segment **303** may have a sinusoidal-shape (e.g., S-profile) including a plurality of convex waves **305** (e.g., peaks) and a plurality of concave waves **307** (e.g., troughs). The plurality of waves **305**, **307** may have various suitable lengths and/or amplitudes relative to a longitudinal axis of flexible body **102**.

(112) In the present example, irregular segment **303** includes a pair of convex waves **305** and a pair of concave waves **307**, however, it should be understood that additional and/or fewer waves **305**, **307** may be included on irregular segment **303**. In other embodiments, flexible body **102** may include additional irregular segments **303** between distal end **104** and proximal end **106**. In the present example, irregular segment **303** may be formed along flexible body **102** adjacent to distal end **104** and may include a constant width with a remaining portion of flexible body **102**. Further, irregular segment **303** may be configured to increase a flexibility of flexible body **102** relative to the remaining portion of flexible body **102**.

(113) Still referring to FIG. **14**, flexible body **102** may include a plurality of electrodes **114** along portions of flexible body **102** proximal to and distal of irregular segment **303** such that irregular segment **303** excludes electrodes **114** thereon. In some embodiments, one or more of convex waves **305** and/or concave waves **307** may include electrodes **114** thereon. In use, irregular segment **303** may be configured and operable to relatively increase the flexural properties of tubular body **12** along the respective portions of tubular body **12** that align with irregular segment **303** when circuit device **300** is attached to medical device **10**.

(114) Referring now to FIG. **15**, an exemplary circuit device **400** is shown in accordance with another example of this disclosure. Circuit device **400** may include one or more angular changes/bends along a longitudinal length of flexible body **102**. For example, flexible body **102** may include a plurality of segments **404**, **406**, **408** separated from one another by at least one bend **410**, respectively. Accordingly, each of the plurality of segments **404**, **406**, **408** may be positioned and aligned relatively transverse to one another.

(115) In the present example, a distal segment **404** is positioned adjacent to distal end **104**, a proximal segment **408** is positioned adjacent to proximal end **106**, and an intermediate segment **406** is disposed between distal segment **404** and proximal segment **408**. Flexible body **102** may include a pair of bends **410**, at least one of which is disposed between distal segment **404** and intermediate segment **406** and another of which is disposed between intermediate segment **406** and proximal segment **408**.

(116) Still referring to FIG. **15**, each of the plurality of segments **404**, **406**, **408** may have varying flexural properties relative to one another based on an angle of bends **410** between an adjacent pair of segments. Bends **410** may range from about 1 degree to about 60 degrees and each bend **410**

may extend an adjacent pair of segments in a different (or similar) direction and/or angle relative to one another. Further, each bend **410** may include varying angles relative to the one or more other bends **410** on flexible body **102**.

(117) In other embodiments, circuit device **400** may include additional and/or fewer segments along flexible body **102** that may be separated by additional and/or fewer bends **410**, respectively. Each of the plurality of segments **404**, **406**, **408** may include one or more electrodes **114**; while in some embodiments, segments **404**, **406**, **408** may omit electrodes **114** entirely.

(118) Referring now to FIG. **16**, each of the plurality of segments **404**, **406**, **408** may cause the corresponding portion of the overall medical device to have varying flexural properties relative to one another based on an angle of helical wrap of each segment **404**, **406**, **408** about tubular body **12**. In other words, the angle, pitch, and/or spacing between windings that flexible body **102** helically winds about tubular body **12** may vary at each segment **404**, **406**, **408**. For example, a pitch of helical wind of flexible body **102** may be relatively greater at distal segment **404** and proximal segment **408** than at intermediate segment **406**. In this instance, flexible body **102** may be substantially flexible along distal segment **404** and proximal segment **408** relative to intermediate segment **406**.

(119) By way of further example, the pitch or angle that flexible body **102** helically winds about tubular body **12** may be relatively smaller along intermediate segment **406** such that flexible body **102** may be substantially rigid along intermediate segment **406**. It should be appreciated that segments **404**, **406**, **408** may include various other angles and pitches of helical wrap than those shown and described herein, and relative to one another, than those shown and described herein dependent on a desired configuration of circuit device **400**.

(120) Still referring to FIG. **16**, a direction of the helical wrap of flexible body **102** may change directions between one or more of the plurality of segments **404**, **406**, **408**. In the present example, distal segment **404** and proximal segment **408** may be wound about tubular body **12** in a first direction A (e.g., clockwise direction) and intermediate segment **406** may wrap around tubular body **12** in a second direction B (e.g., counter clockwise direction) that is different than the first direction A. Further, distal segment **404** and/or proximal segment **408** may helically wind about tubular body **12** at a first helical pitch α and intermediate segment **406** may helically wind about tubular body **12** at a second helical pitch β that is different than first helical pitch α . For example, first helical pitch α may be relatively less than second helical pitch β such that distal segment **404** and/or proximal segment **408** may include fewer helical winds than intermediate segment **406**. It should be appreciated that segments **404**, **406**, **408** may include various other helical pitches and/or winds relative to one another than those shown and described herein.

(121) Circuit device **400** may be configured to form a gap **405** between adjacent segments **404**, **406**, **408** that change direction relative to one another when flexible body **102** is attached to tubular body **12**. Circuit device **400** may not cover at least a portion of tubular body **12** when the helical wrap of flexible body **102** changes directions, thereby exposing a portion of tubular body **12** disposed beneath flexible body **102**.

(122) It should be appreciated that the flexural properties of tubular body **12** may vary along the portions that are exposed along gaps **405** relative to other portions that receive flexible body **102**. For example, tubular body **12** may have a relatively decreased rigidity (i.e., an increased flexibility) along regions that omit flexible body **102** thereon. Further, an assembly of medical device **10** and circuit device **100** may provide areas having a greater material thickness along portions where flexible body **102** is attached to tubular body **12**, thereby providing asymmetric properties along the longitudinal length of tubular body **12**.

(123) Still referring to FIG. **16**, one or more features of medical device **10** on tubular body **12** (e.g., access holes **18**) may be revealed and accessible along gap **405** when flexible body **102** is wound about tubular body **12**. A size of gap **405** may vary and may be dependent on, for example, the angle of the helical wrap between the adjacent pair of segments defining gap **405**. Access holes **18**

may be in fluid communication with one or more lumens **11** (FIG. 6) such that a fluid and/or a device (e.g. a guidewire) may be delivered outwardly from tubular body **12** and/or a received into tubular body **12** via access holes **18**. In some examples, access holes **18** may provide sensing portals to allow a device (e.g., fiber optic sensor, etc.) to measure one or more parameters, such as, for example, oxygen levels of a fluid (e.g., blood, etc.), central venous pressure, and more. Further, access holes **18** may provide deployment portals that allow one or more instruments to extend externally from tubular body **12**. Although not shown, in some embodiments, access holes **18** may facilitate deployment of a Nitinol positioning device from tubular body **12** for contacting a vessel wall, thereby biasing tubular body **12** to a preferred configuration and stabilizing medical device **10** relative to a target treatment site.

(124) It should be appreciated that an angle and/or a pitch of helical wrap of each the plurality of segments **404**, **406**, **408** may remain constant and/or vary relative to one another when a direction of the helical wind changes. In some embodiments, bend **410** may match an angle of the helical wrap of flexible body **102**. In this instance, at least one of the plurality of segments **404**, **406**, **408** may have an elongated length extending along a longitudinal axis that is transverse to the remaining segments of flexible body **102**.

(125) For example, as seen in FIG. 17, distal segment **404** may extend parallel to a central axis of tubular body **12** when circuit device **400** is secured to medical device **10**, and an angular change at bend **410** between distal segment **404** and intermediate segment **406** may be determinative of an angle of the helical wrap of flexible body **102** about tubular body **12**. Stated differently, an angle of bend **410** may define, or correspond to, a helix angle of flexible body **102** wrapping about the central axis of tubular body **12**.

(126) As seen in FIG. 17A, bend **410** may define a bend angle C between distal segment **404** of flexible body **102** and intermediate segment **406**. In some embodiments, bend angle C may range from about 1 degree to about 179 degrees, and may at least partially define a configuration of the plurality of helical winds (of flexible body **102**) relative to tubular body **12**. For example, relative to an axis T that is transverse to a central axis L of tubular body **12**, the plurality of helical winds along intermediate segment **406** may include a helix angle D. Helix angle D may have a relationship relative to bend angle C such that bend angle C may be substantially equal to helix angle D plus about 90 degrees, when distal segment **404** is aligned with, or substantially parallel to, axis of tubular body **12**, ($C=D+90$ degrees).

(127) It should be understood that bend angle C and helix angle D of the present example are merely illustrative such that bend **410** and the plurality of helical winds along intermediate segment **406** may have various other suitable relationships relative to one another than those shown and described herein. Additionally, a helical pitch of intermediate segment **406** (e.g., see FIG. 16, first helical α , second helical pitch β , etc.) may be at least partially defined by, or otherwise correspond to, bend angle C and/or helix angle D of flexible body **102**. Thus, intermediate segment **406** may be wrapped in a helical configuration about tubular body **12**. In the present example, circuit device **400** only includes a single bend **410** such that intermediate segment **406** extends to and terminates at bend **108**.

(128) Flexible body **102**, and the portion of the medical device corresponding to it, may have varying flexural properties along distal segment **404** and intermediate segment **406** based on a respective geometry of each segment **404**, **406**. In the present example, distal segment **404** may be oriented parallel to a central axis of tubular body **12** such that the portion of the medical device corresponding to it distal segment **104** may have radial asymmetric flexural properties. Medical device **10** may experience greater deflection along a portion corresponding to distal segment **404** when a force that is transverse to a longitudinal axis of flexible body **102** and parallel to a central axis of tubular body **12** is applied, than when the force is perpendicular to both flexible body **102** and tubular body **12**.

(129) Still referring to FIG. 17, a radial flexural asymmetry of distal segment **404** may provide a

greater rigidity on tubular body **12**, such as, for example, in a specific direction when tubular body **12** is bent during use in a procedure. For example, a distal portion of tubular body **12** may be bent in one of a plurality of directions relative to a target treatment site, such as, for example, to contact a vessel wall within a subject. The distal portion of tubular body **12** may maintain contact with the vessel wall due to an enhanced rigidity of the distal portion provided by the radial flexural asymmetry of distal segment **404**. In contrast to the radial asymmetric flexural properties of distal segment **404**, intermediate segment **406** may have radial symmetric flexural properties based on the geometry of intermediate segment **406**.

(130) Referring now to FIG. **18**, another exemplary circuit device **500** is shown in accordance with an example of this disclosure. Circuit device **500** may include one or more segments **504** having a stiffening structure formed therein, and one or more segments **506** omitting a stiffening structure. In the present example, circuit device **500** may include a pair of stiffened segments **504** and a pair non-stiffened segments **506** positioned between distal end **104** and proximal end **106**. Stiffened segments **504** alternate with non-stiffened segments **506** along the length of circuit device **500**. In other embodiments, circuit device **500** may include additional and/or fewer segments **504**, **506** along flexible body **102** in various other suitable arrangements than those shown and described herein.

(131) The stiffening structures may include components (e.g., additional layers, adhesive used for lamination, stiffening wires, etc.) made of materials of varying stiffness and/or thickness and that may be incorporated into regions of flexible body **102** to increase and/or decrease a stiffness of circuit device **500**. Stiffeners may include glass-reinforced epoxy laminate materials (FR4), polyimide, polyester, polymers, and/or stainless steel. The stiffening structures may be formed of materials having a relatively high stiffness to increase a rigidity of tubular body **12** in the specific areas aligned with stiffening segments **504** when flexible body **102** is attached thereto.

(132) Stated differently, flexible body **102** may be configured to reduce a flexibility of tubular body **12** along areas aligned with stiffening segments **504**. The stiffening structures may provide additional layers on tubular body **12**, thereby varying a width of the assembly of medical device **10** and circuit device **500** along a longitudinal length of tubular body **12**. Accordingly, the flexural properties of tubular body **12** may vary at the selective regions corresponding to a location of stiffened segments **504** thereon. In other embodiments, a stiffening structure may be disposed between an exterior surface of tubular body **12** and interior surface **103** of flexible body **102** prior to attaching circuit device **100** to medical device **10**. In this instance, the stiffening structure may be encapsulated between tubular body **12** and flexible body **102**.

(133) Referring now to FIG. **19**, another exemplary circuit device **600** is shown in accordance with an example of this disclosure. Circuit device **600** may include a distal segment **604** and an intermediate segment **606** defining a longitudinal length of flexible body **102** between distal end **104** and proximal end **106**. Distal segment **604** may include a pair of branch segments **608** extending distally from intermediate segment **606** such that distal end **104** may be defined by the pair of branch segments **608**. Each branch segment **608** may be in parallel alignment with one another and may have a longitudinal length that is equal to, greater than, and/or less than one another. In the present example, branch segments **608** may have a substantially similar length that is less than a longitudinal length of intermediate segment **606**. Each of branch segments **608** may have a width that is relatively less than a width of intermediate segment **606**.

(134) Still referring to FIG. **19**, flexible body **102** may further include a bend **610** at a proximal end of the pair of branch segments **608** such that branch segments **608** may be angled relative to intermediate segment **606**. Thus, each branch segment **608** may have a longitudinal axis that is transverse to an axis of intermediate segment **606**. In some examples, an angular change of flexible body **102** at bend **610** may be determinative of an angle of a helical wrap of intermediate segment **606** about tubular body **12** when circuit device **600** is secured to medical device **10** (see, e.g., FIG. **17A**, bend angle C).

(135) Distal segment **604** and/or intermediate segment **606** may include one or more (e.g., a plurality) of electrodes **114**. In the present example, each of the pair of branch segments **608** may define an array of electrodes **114** along distal segment **604**, thereby providing a complex circuit density at distal segment **604**. It should be appreciated that circuit device **600** may be configured to generate greater current and/or fields of coverage with the array of electrodes **114** on branch segments **608** for providing enhanced treatment therapies to a target site. The array of electrodes **114** may provide medical device **10** with complex electrode geometries capable of selectively generating greater current densities and/or concentrated density fields.

(136) Circuit device **600** may be assembled onto medical device **10** such that the longitudinal axis of each branch segment **608** may be parallel to a center, longitudinal axis of tubular body **12**. Stated differently, distal segment **604** may be secured onto a distal portion of tubular body **12** in a linear arrangement such that branch segments **608** extend parallel to the center axis of tubular body **12**. Further, an axis of intermediate segment **606** may be transverse to the center axis of tubular body **12** such that flexible body **102** may be helically wound about an exterior of tubular body **12** along intermediate segment **606**. Accordingly, circuit device **600** may be configured to impart different flexural properties onto tubular body **12** along a portion receiving distal segment **604** and intermediate segment **606**, respectively.

(137) Still referring to FIG. **19**, branch segments **608** may be spaced apart from one another by an angle and/or distance when flexible body **102** is mounted onto tubular body **12**. For example, each branch segment **608** may be positioned about 180 degrees apart from another when secured to a distal portion of tubular body **12**. Thus, circuit device **600** may be configured to provide a pair of opposing electrode arrays that are offset from one another and positioned about a circumference of tubular body **12**. For example, branch segments **608** may be positioned along a top and bottom (e.g., or a left and right) exterior surface of tubular body **12** when circuit device **600** is attached to medical device **10**.

(138) It should be appreciated that branch segments **608** may be selectively spaced apart from one another at various other relative degrees and/or distances dependent on a desired therapeutic effect to be provided by circuit device **600** and medical device **10**. In other embodiments, circuit device **600** may include additional branch segments **608** at distal segment **604**, intermediate segment **606**, and/or at various other portions of flexible body **102**.

(139) For example, as shown in FIG. **20**, circuit device **600** may include a second pair of branch segments **609** extending distally from intermediate section **606** and angled relative thereto by bend **610**. In the present example, the pair of branch segments **609** may extend parallel to branch segments **608** and may have a longitudinal length that is equal to, greater than, and/or less than a longitudinal length of branch segments **608**. In the example, branch segments **609** may have a smaller length than branch segments **608** and may be positioned in an alternating configuration with branch segments **608**. Thus, each branch segment **608** may be separated from one another by at least one branch segment **609** positioned therebetween.

(140) Each branch segment **609** may include one or more (e.g., a plurality) of electrodes **114**, thereby defining an array of electrodes **114** on each branch segment **609**. In some examples, electrodes **114** may be arranged along each branch segment **609** to be longitudinally aligned with one another, and longitudinally offset from the plurality of electrodes **114** along branch segments **608**.

(141) Still referring to FIG. **20**, branch segments **609** may be spaced apart from one another, and/or from branch segments **608**, by an angle and/or distance when flexible body **102** is mounted onto tubular body **12**. For example, each branch segment **609** may be positioned about 180 degrees from one another and about 90 degrees from each branch segment **608** when secured to a distal portion of tubular body **12**. Thus, circuit device **600** may provide two pairs of opposing electrode arrays that are offset from one another and positioned about a circumference of tubular body **12**.

(142) For example, FIG. **21** shows branch segments **608** positioned along a top and bottom exterior

surface of tubular body **12** (only the top can be seen in the Figure), and branch segments **609** positioned along a left and right side surface (only the left side can be seen in the Figure) when circuit device **600** is attached to medical device **10**. It should be understood that branch segments **608** are spaced apart from one another by 180 degrees on tubular body **12** and branch segments **609** are spaced apart from one another by 180 degrees. Further, each branch segment **608** is positioned apart from branch segment **609** by 90 degrees on tubular body **12**. In other embodiments, circuit device **600** may include additional and/or fewer branch segments **608**, **609** at distal segment **604**, intermediate segment **606**, and/or at various other portions of flexible body **102**.

(143) Referring now to FIG. 22, an exemplary circuit device **700** is schematically depicted in accordance with another example of this disclosure. Circuit device **700** may include flexible body **102** having a longitudinal length defined by a distal segment **702**, a first intermediate segment **704**, a second intermediate segment **706**, and a proximal segment **708**. Distal segment **702** may terminate at distal end **104** and extend distally relative to first intermediate segment **704**. Distal segment **702** may include the pair of branch segments **608** extending at an angle relative to first intermediate segment **702** by a bend **710**.

(144) First intermediate segment **704** may extend distally relative to second intermediate segment **706** such that first intermediate segment **704** may be disposed between distal segment **702** and second intermediate segment **706**. First intermediate segment **704** may have a longitudinal length (an axis) extending transverse to the longitudinal axes of each of distal segment **702** and second intermediate segment **706**. In some examples, first intermediate segment **704** may include one or more (e.g., a plurality) of electrodes **114**; while in other examples, first intermediate segment **704** may exclude electrodes **114** thereon.

(145) Still referring to FIG. 22, flexible body **102** may further include a second bend **712** between first intermediate segment **704** and second intermediate segment **706** such that first intermediate segment **704** may be configured to extend at an angle relative to second intermediate segment **706** by bend **712**. Second intermediate segment **706** may include the pair of branch segments **609**, with first intermediate segment **704** extending distally from a distal end of at least one of the pair of branch segments **609**. Stated differently, each branch segment **608**, **609** has a free distal end except for at least one of branch segments **609** which is secured to a proximal end of first intermediate segment **704**.

(146) Accordingly, the pair of branch segments **609** may be positioned proximal to, and at angle from, the pair of branch segments **608**. Thus, each branch segment **609** may have a longitudinal axis that is transverse to an axis of the pair of branch segments **608**. Second intermediate segment **706** may extend distally relative to proximal intermediate segment **708** such that second intermediate segment **706** may be disposed between first intermediate segment **704** and proximal segment **708**. Second intermediate segment **706** may have a longitudinal length (an axis) extending transverse to a longitudinal axis of each of first intermediate segment **704** and proximal segment **708**. In some examples, first intermediate segment **704** may have an axis that is parallel to an axis of proximal segment **708**.

(147) Still referring to FIG. 22, second intermediate segment **706** may include one or more (e.g., a plurality) of electrodes **114** on one or more of the pair of branch segments **609**; while in other examples, second intermediate segment **706** may exclude electrodes **114** entirely. Flexible body **102** may include a bend **714** between second intermediate segment **706** and proximal segment **708** such that second intermediate segment **706** may be configured to extend at an angle relative to proximal segment **708** by bend **714**.

(148) Distal segment **702** and second intermediate segment **706** may extend parallel to each other and to a central longitudinal axis of tubular body **12**, and first intermediate segment **704** and proximal segment **708** may be helically wrapped about tubular body **12**, when circuit device **700** is attached to medical device **10**. Accordingly, varying flexural properties may be imparted on tubular body **12** along those portions receiving segments **702**, **706** thereon compared to segments **704**, **708**.

(149) The pair of branches **608** may be secured to tubular body **12** along a similar or different radial location about the axis of body **12** than the pair of branches **609** (i.e., either spaced 180 degrees or 90 degrees apart, respectively, or any other suitable angle apart). An angular change of flexible body **102** at bends **710**, **712**, **714** may be determinative of an angle of a helical wrap of first intermediate segment **704** and proximal segment **708** about tubular body **12** when circuit device **700** is secured to medical device **10** (see, e.g., FIG. 17A, bend angle C). Accordingly, one or more gaps may be formed between branch segments **608** and branch segments **609**.

(150) Referring now to FIG. 23, an exemplary circuit device **800** is schematically depicted in accordance with another example of this disclosure. Circuit device **800** may include an alternative exemplary proximal tab **110''** at proximal end **106**. Proximal tab **110''** may include a rectangular shape having one or more (e.g. a plurality) conductive pads **112** positioned thereon. In the example, proximal tab **110''** may define a connector interface of circuit device **800** for connection with a power source (not shown). Stated differently, proximal tab **110''** may be configured to directly connect with a power source in lieu of the circuit device establishing connection with the power source via one or more wires **118** (FIGS. 8-9).

(151) FIG. 24 shows electrical connector **156** in a disassembled state for illustrative purposes. As shown, flexible body **102** may be received within electrical connector **156** via protective sheath **154**. In the example, protective sheath **154** may be secured to electrical connector **156** at an inlet port **162** of electrical connector **156**. Further, electrical connector **156** may include a top housing **160A** and a bottom housing **160B** defining an interior cavity for receiving proximal end **106**. In the example, electrical connector **156** may include an outlet connector **164** in lieu of socket **158** and pins **159** (see FIG. 11).

(152) Outlet connector **164** may extend proximally outward from housings **160A**, **160B** when electrical connector **156** is in a fully assembled state. Outlet connector **164** may define a connector interface of electrical connector **156** for connection with a power source and may be configured to receive proximal tab **110''**. Proximal tab **110''** may be disposed about an exterior of outlet connector **164** with the plurality of conductive pads **112** facing radially outward. Outlet connector **164** may include a keying feature **166** configured to facilitate an alignment and attachment of proximal tab **110''** to outlet connector **164**. For example, keying feature **166** may engage the terminal ends of proximal tab **110''** and/or designate a starting and ending point for wrapping proximal tab **110''** about a circumference of outlet connector **164**.

(153) Still referring to FIG. 24, proximal tab **110''** may be secured to outlet connector **164** by various suitable mechanisms, including, for example, by an adhesive. In exemplary use, conductive pads **112** may be exposed from housings **160A**, **160B** when proximal end **106** is received within electrical connector **156** and proximal tab **110''** is secured to outlet connector **164**. Conductive pads **112** may connect directly to the power source when electrical connector **156** is coupled thereto and outlet connector **164** is received within the power source, thereby providing electrical power to circuit device **800**.

(154) Each of the aforementioned systems, devices, assemblies, and methods may be used to provide electrical circuitry at a target treatment site in a subject (e.g., a patient) that may be capable of providing diagnostic or therapeutic effects. By providing a medical device including a circuit device disposed at the exterior of the medical device (over the sheath or tube), a user may utilize the lumens of the medical device for receipt of others devices and/or materials. With the circuit device having a minimal size and flexible electrical circuitry, the circuit device may minimize an overall physical profile of the medical device while maximizing a selective positioning of the electrical components at various locations along the medical device. In this instance, a user may increase efficiency of procedures, reduce overall procedure time, and/or avoid unnecessary harm to a subject's body caused by the medical device having an invasive profile for accommodating electrical circuitry devices within the lumens.

(155) It will be apparent to those skilled in the art that various modifications and variations may be

made in the disclosed devices and methods without departing from the scope of this disclosure. Other aspects of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the features disclosed herein. It is intended that the specification and examples be considered as exemplary only.

Claims

1. A medical device comprising: a tube having a longitudinal length and at least one lumen; a flexible body including a stimulation array configured to provide an electrical charge for stimulating tissue; an electrical connection positioned on the flexible body proximal to the stimulation array, wherein the electrical connection is configured to electrically couple the stimulation array to a power source; and an electronic component positioned between the stimulation array and the power source, wherein the electronic component includes a memory and a processor configured to implement programmable code stored on the memory; wherein the flexible body is configured to cover at least a portion of the longitudinal length of the tube such that the stimulation array is disposed about an exterior of the tube.
2. The medical device of claim 1, wherein the memory is configured to store data relating to a serial number, a lot number, a product configuration, an expiration date, a calibration constant, sensor data, or a combination thereof.
3. The medical device of claim 1, wherein the memory is an electrically erasable programmable read-only memory.
4. The medical device of claim 1, wherein the flexible body includes (1) a lead that electrically couples the electrical connection to the stimulation array and (2) a dielectric substrate film between the lead and the stimulation array.
5. The medical device of claim 4, wherein the flexible body further includes a sensor disposed within the dielectric substrate film; and the processor of the electronic component is configured to: receive information from the sensor; and process the information received from the sensor to produce a control signal for controlling operation of the medical device.
6. The medical device of claim 1, wherein the flexible body is helically wound about the exterior of the tube in a plurality of winds, and a first position of the tube that contacts the helically wound flexible body has a different flexibility than a second position of the tube that contacts the helically wound flexible body.
7. A medical device comprising: a flexible body; an electrode positioned on the flexible body, wherein the electrode is configured to provide an electrical charge for stimulating tissue; an electrical connection including a proximal end and a distal end, wherein the electrical connection is configured to electrically couple the electrode to a power source; an electrical lead connecting the electrode to the electrical connection, wherein the electrical lead is on or in the flexible body; an electronic component between the proximal end of the electrical connection and the electrode, wherein the electronic component includes a memory, a sensing circuit, a radio-frequency identification tag, a processor, or a combination thereof; and a dielectric substrate film disposed between the electrode and the electrical lead, wherein the dielectric substrate film includes a hole filled with a conductive material electrically connecting the electrode to the electrical lead.
8. The medical device of claim 7, wherein the electronic component includes a memory, the memory is an electrically erasable programmable read-only memory (EEPROM), and the EEPROM is configured to store data related to an expiration date, a date of activation, and/or data from a therapy session.
9. The medical device of claim 7, wherein the distal end of the electrical connection is proximate to a proximal end of the flexible body.
10. The medical device of claim 7, wherein the electronic component is mounted on the flexible body.

11. The medical device of claim 10, further comprising a hub, wherein the electronic component is located within the hub.
 12. The medical device of claim 11, wherein the hub further includes an inlet, and the flexible body passes through the inlet.
 13. The medical device of claim 7, further including a sensor, an electronic chip, or an integrated circuit disposed within the dielectric substrate film and between the electrode and the electrical lead.
 14. A medical device comprising: a tube including a lumen; and a circuit having a flexible body, wherein the circuit comprises: a distal portion, wherein the distal portion includes an electrode configured to provide an electrical charge for stimulating tissue; a proximal portion coupled to a distal end of an electrical connection, wherein the proximal portion is angled relative to the distal portion, and the electrical connection is configured to electrically couple the electrode to a power source; an electronic component between a proximal end of the electrical connection and the distal portion of the circuit, wherein the electronic component includes a memory, a sensing circuit, a radio-frequency identification tag, a processor, or a combination thereof; a conductive lead electrically coupled to the electrical connection and the electrode; and a via between the conductive lead and the electrode, wherein the via provides an electrical connection between the conductive lead and the electrode.
 15. The medical device of claim 14, wherein the electronic component is configured to record use data of the medical device.
 16. The medical device of claim 14, wherein a length of the circuit is helically wound around the tube.
 17. The medical device of claim 14, wherein the circuit includes at least one segment having an enlarged width relative to an adjacent portion of the circuit, and wherein the at least one segment is less flexible than the adjacent portion of the circuit.
 18. The medical device of claim 14, wherein the medical device further includes a hub comprising: an inlet, wherein a length of the circuit is positioned within the inlet; and an extension tube configured to pass fluids to and from the lumen.
 19. The medical device of claim 18, wherein the electrical connection is positioned within the hub.
 20. The medical device of claim 14, wherein: the electrode is a first electrode; the distal portion of the circuit includes a first branch and a second branch; the first and second branches are arranged substantially parallel to a longitudinal axis of the tube; the first branch includes the first electrode and a second electrode; and the second branch includes a third electrode and a fourth electrode.
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